

Pokhara University
Faculty of Science and Technology

Course Code: MTH 210
Course title: Calculus II (3-2-0)
Nature of the Course: Theory
Level: Bachelor

Full Marks: 100
Pass Marks: 45
Total Lectures: 45 hours
Program: BE

1. Course Description

The Calculus II is designed to develop the competency of the students in the applications of various mathematical concepts they learned in previous semesters. It is mainly equipped with Vector Calculus, Laplace transform, Multiple integrals, Differential Equations, Fourier Series and with introduction of Partial differential equations. The pre-requisite for this course is Calculus I and Algebra & Geometry. The course will be delivered through lecture method, assignments on practically based engineering problems and class tests.

2. General Objectives

The course is designed to acquaint the students with applications of mathematics in engineering.

3. Methods of Instruction

Lecture, tutorials, discussions and assignments

4. Contents in Detail

Specific objectives	Contents
Evaluate multiple integrals	Unit I: Multiple Integrals (6 Hours) 1.1 Introduction 1.2 Double integrals in Cartesian and polar form, Fubini's theorem (statement only), change of order of integration, change of variable from in double integral Jacobian matrix and reduction into Polar. 1.3 Triple integrals in Cartesian form and Dirichlet's Integral, use of cylindrical and spherical coordinates to evaluate triple integral. 1.4 Application of double and triple integrals to find Area and volume.
Analyze the concept of solution of differential equations in terms of infinite series as power series.	Unit II: Series Solution of Differential Equations and Special Functions (6 Hours) 2.1 Power series method of solution of differential equations. 2.2 Legendre's Equation, Legendre's polynomials $P_n(x)$ of x . Graph of $P_0(x)$, $P_1(x)$, $P_2(x)$. 2.3 Frobenius method. Bessel's equation, Bessel's function $J_n(x)$ and its properties. Graph of $J_0(x)$ for $x = 1$ to 2

Apply the integral transform in solving practical problems	Unit III: Laplace Transform and Its Application (8 Hours) 3.1 Laplace Transform (LT), Inverse LT, Linearity of LT, LT of elementary functions, inverses and first shifting (sshifting) theorem. Existence theorem of Laplace transform (without proof) and uniqueness. 3.2 Transform of Derivative and Integrals of a function. 3.3 Differentiation and Integration of Laplace transform.
	3.4 Unit step function, periodic function and LT, second shifting (t-shifting) theorem. 3.5 Convolution theorem and its application to find inverse. 3.6 Application of Laplace transform to find the solutions of ordinary differential equations (IVP).
Solve higher dimensional (multivariable) calculus problems	Unit IV: Advanced Vector Calculus (15 Hours) 4.1 Differentiation of vector function of scalar variable. 4.2 Point functions, Gradient, directional derivative, divergence and curl with properties (without proof) 4.3 Line integral with physical interpretation and evaluation of line integrals on various path 4.4 Line integral, potential function and independence of path 4.5 Green's theorem in plane (without proof) and its various applications 4.6 Surface integral and evaluation of surface integrals 4.7 Stoke's theorem (without proof) and its applications 4.8 Gauss Divergence theorem (without proof) and its applications.
Illustrate periodic functions of practical importance by infinite trigonometric series	Unit V: Fourier Series (5 Hours) 5.1 Periodic Functions, odd and even functions 5.2 Fourier series of 2π periodic functions in the interval $(\pi, \pi + 2\pi)$. 5.3 Fourier series of 2π periodic functions. 5.4 Fourier series of odd and even functions, sine and cosine series
Interpret physical phenomenon by partial differential equations	Unit VII: Partial Differential Equations (5 Hours) 7.1 Introduction 7.2 Linear constant coefficient equation 7.3 Applications in conservation laws, the breaking time, shock waves, nonlinear advection equations, and traffic flow.

Note: The figures in the parentheses indicate the approximate periods for the respective units.

5. List of Tutorials (30 hours)

Tutorial work covers the work to be done in tutorial. This will enable the students to compute the mathematical problems under the supervision of the course leader. The major tutorial works are as follows:

Unit	Unit name	List of Tutorials	Tutorial hours
1	Unit I: Multiple Integrals	1.1 Problems on double integral by changing order of integration and reduction into polar. 1.2 Triple integral with examples on Dirichlet's integrals, use Cylindrical and Spherical coordinates. 1.3 Problems on area and volume by double and triple integral	2 hrs 1 hr 1 hr
2	Unit II: Series Solution of Differential	2.1 Solve Legendre's polynomials $P_n(x)$ of different order. 2.2 Solve Bessel's function $J_n(x)$ and their properties.	2 hrs 2 hrs
	Equations and Special Functions		
3	Unit III: Laplace Transform and Its Application	3.1 Problems on Laplace and Inverse Laplace transform of different functions. 3.2 Solution of IVP using Laplace transform.	2 hrs 2 hrs
4	Unit IV: Advanced Vector Calculus	4.1 Problems on gradient, Normal vector to a surface, Directional derivative, angle between two surfaces. 4.2 Problems on divergence and curl. 4.3 Problems on line integrals, Exactness and path independence. 4.4 Problems based on Green's theorem, Stoke's theorem and Gauss divergence theorem.	2 hrs 2 hrs 3 hrs 3 hrs
5	Unit V: Fourier Series	5.1 Problems on Fourier series in $(-\pi, \pi)$, $(0, 2\pi)$ and $[-\pi, \pi]$. 5.2 Problems on Fourier series of odd and even functions, Fourier series in general interval $(-\pi, \pi)$.	2 hrs 2 hrs
6	Unit VII: Partial Differential equation	6.1 Solve partial differential equations and their applications.	4 hrs

6. Evaluation System and Students' Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows.

Internal Evaluation	Marks	External Evaluation	Weight	Marks
Attendance & Class Participation	10%	Semester End Board Examination	50%	50
Assignments	20%			
Presentations/Quizzes	10%			
Term exam	60%			
Total Internal	50			
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

Text Books

1. Kreyszig, E. Advance Engineering Mathematics, New Delhi: John Wiley and Sons Inc.
2. Stewart, J. Calculus, Early Transcendental. India; Cengage Learning.

References

1. Dass, H. K. & Verma R. Higher Engineering Mathematics. New Delhi: S Chand Publishing.
2. Mishra, P., Mishra, R., Mishra, V. P., & Mishra, M. Advance Engineering Mathematics. New Delhi: V. P. Mishra Publication.
3. Thomas, G. & Finney, R. Calculus and Analytical Geometry. New Delhi: Narosa Publishing House.

Course Code: CMP 234

Course title: Computer Graphics (3-1-2)

Nature of the Course: Theory and Practical

Level: Bachelor

Full Marks: 100

Pass Marks: 45

Total Lectures: 45 hours

Program: BE

1. Course Description

This course intends to provide students with an in-depth knowledge of a machine's perception of visual information for storing, accessing and manipulating visual representations of graphical objects and scenes. The course contents cover theoretical, practical, mathematical and algorithmic aspects of concepts such as graphical hardware, two dimensional and three dimensional representations of graphical objects, the use of techniques such as projection, visible surface detection and lightning effects. The main aim of the course is also to impart knowledge to students regarding the techniques used for incorporating visual realism in the computer-generated images and scenes portrayed on a display device.

2. General Objectives

The course is designed with the following objectives:

- To make the students familiar with the technologies used for representing and rendering of graphical objects and scenes in a computer system
- To equip students with necessary skills and knowledge of Two dimensional and Three dimensional graphics necessary for viewing and performing transformations
- To provide students with a good amount of knowledge of algorithms and approaches used to incorporate maximum amount of visual realism in the computer generated graphical scenes and objects.
- To familiarize students with the graphical standards used for building platform independent portable software

3. Methods of Instruction

As the course consists of theoretical, mathematical and practical aspects of Computer Graphics, the delivery needs to include lecture, tutorial, practical classes. Apart from that, in order to share knowledge and assimilate new ideas and emerging trends, students can be involved in group discussions and presentations pertaining to the field of Computer Graphics. Short quizzes can be held in the class to check the students' level of comprehension. Project work can be assigned to students to come up with a graphical software product that demonstrates their level of understanding of the contents studied in the course.

4. Contents in detail with specific objectives

Specific Objectives	Contents
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Provides a brief introduction of the field of Computer Graphics and familiarize students with basic graphics hardware and software. It intends to enhance the understanding of the technology used for producing graphical output on display devices.	Unit 1: Overview of Computer Graphics and Graphics Systems (7 hrs) 1.1 Introduction, Applications and Recent trends of Computer Graphics 1.2 Interactive Input Devices 1.3 Display Devices, Color Monitors, Hard Copy Devices 1.4 Raster and Random Scan Systems and Architectures 1.5 Video Controller
	1.6 Use of Digital to Analog Converter and Frame Buffer Organization 1.7 Graphics Software, Modern Graphics Hardware (GPU)
Learn about various scan conversion and area filling algorithms. Students will be able to implement these scan conversion algorithms using a programming language. This chapter helps students learn how algorithms are used to generate output primitives and fill them with specified intensities.	Unit 2: Scan Conversion Algorithms (7 hrs) 2.1 Scan Conversion 2.2 Line Drawing Algorithms 2.2.1 Digital Differential Analyzer 2.3.2 Bresenham's Algorithm 2.3 Circle Generation Algorithm 2.4 Ellipse Generation Algorithm 2.5 Filled Area Primitives 2.5.1 Scan Line Polygon Fill Algorithm 2.5.2 Boundary Fill Algorithm, Flood Fill Algorithm
Get students well acquainted with the two-dimensional geometric transformations and viewing Able to make use of various transformation operations and viewing operations to reposition and resize objects in two dimensional scene Learn about the operations used in viewing routines that convert a world coordinate scene description to a display for an output device.	Unit 3: Two Dimensional Geometric Transformations and Viewing (8 hrs) 3.1 Two Dimensional Transformations 3.1.1 Translation, Rotation, Scaling, Shearing, Reflection 3.2 Matrix Representations of Transformations 3.3 Homogeneous Coordinate System 3.4 Composite Transformations 3.5 Windowing Concepts, Two Dimensional Viewing Pipeline 3.6 Window to Viewport Transformation 3.7 Line Clipping Algorithm: Cohen-Sutherland 3.8 Polygon Clipping: Sutherland-Hodgeman

Learn about the extensions of twodimensional methods for geometric transformation in three dimensions. Helps them apply the knowledge of three dimensional rendering and transformations for representing objects more accurately	Unit 4: Graphics in Three Dimensions (8 hrs) 4.1 Three Dimensional Coordinate Systems 4.2 Three Dimensional Transformations 4.2.1 Translation, Rotation, Scaling, Reflection 4.3 Three Dimensional Representations 4.3.1 Polygon Surfaces 4.3.2 Cubic Spline and Beizer Curve 4.3.3 Non-Planer Surface: Bezier Surface 4.3.4 Fractal Geometry Method 4.4 Three Dimensional Viewing Transformation and Pipeline 4.5 Projections 4.5.1 Parallel Projection 4.5.2 Perspective Projection 4.6 Clipping in Three Dimensions
Get acquainted with the techniques used for bringing visual realism in computer generated graphical output primitives.	Unit 5: Visual Realism (7 hrs) 5.1 Hidden Surfaces and Hidden Surface Removal Approaches 5.1.1 Back-Face Detection 5.1.2 Depth Buffer Method 5.1.3 A- buffer method
Familiarize with visible surface detection techniques for creating realistic displays and in computing light intensities of surfaces using lighting or shading models for generating realistic views	5.1.4 Scan Line Method 5.1.5 Depth Sorting Method 5.2 Illumination and Shading Methods 5.2.1 Illumination Theory and Models 5.2.1.1 Ambient Light 5.2.1.2 Diffuse Reflection 5.2.1.3 Specular Reflection (Phong Model) 5.2.2 Polygon Surface Shading Methods 5.2.2.1 Constant Shading (Flat Shading) 5.2.2.2 Gouraud Shading 5.2.2.3 Phong Shading 5.2.2.4 Fast Phong Shading 5.3 Color Models 5.3.1 RGB 5.3.2 CMYK
Gain a good understanding of the use of graphical standards necessary for achieving software portability. Become familiar with the basic structure or format of a graphical file and know the ways of visually representing data sets. Learn about writing graphical programs that can be ported to multiple hardware platforms without extensive rewriting of the programs through the use of graphical standards.	Unit 6: Graphical Standards (4 Hrs) 6.1 Need for Machine Independent Graphical Languages 6.2 Graphical Standards: PHIGS, GKS 6.3 Graphics Software Standards and Language Binding 6.4 Overview of Graphics File Formats 6.5 Visualization of Data Sets

Learn about the library functions provided in OpenGL for graphical rendering and performing geometric transformations. Enable students to write programs using functions in OpenGL library for drawing graphical output primitives and performing basic transformations	Unit 7: Introduction to OpenGL (4 Hrs) 7.1 Overview and Basic Architecture of OpenGL 7.2 GL Related Libraries 7.2.1 Drawing Basic Output Primitives and Polygons 7.2.2 Call Back Functions and Input Handling 7.2.3 Basic Transformations 7.2.4 Projections and Lighting

5. List of Tutorials

The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover all the required contents of this course.

S.N.	Tutorials
1	Solving problems related to Raster Graphics system, e.g. computation of frame buffer size, color manipulation techniques, aspect ratio, refresh rate, resolution etc
2	Identifying the points for digitizing line, circle and ellipse using the algorithms studied
3	Solving problems related to Two Dimensional Transformations and Matrix Compositions like fixed point scaling, pivot point rotation, reflection about arbitrary line etc
4	Mapping of object descriptions from window to viewport and clipping of line segments against a clip window
5	Solving problems related to Three Dimensional Transformations
6	Computing points necessary for constructing a Bezier Curve with the given set of control points and number of line segments
7	Solving problems related to Parallel and Perspective Projection

9	Determining surface normal of polygons and determining visibility using Back face detection approach
10	Determining averaged intensity at a point of a polygon using Gouraud Shading

6. Practical Work

Students are required to implement the 2D and 3D graphics algorithms studied in the course using C/C++ and OpenGL.

Students are required to demonstrate the assimilated knowledge in Computer Graphics through project work (Games, Graphical Simulations) submitted at the end of the semester. Students need to form a team of 4-5 members and they are allowed to explore new programming languages or platforms (Unity, Unreal Engine or any related tools) for accomplishing the project work.

S.N.	Practical works
1	To familiarize students with the tools to be used for implementing algorithms studied in the course
2	To draw a straight line using Digital Differential Analyzer and Bresenham's Line Drawing Algorithm
3	To digitize a circle using Circle Drawing Algorithm
4	To digitize an ellipse using Ellipse Drawing Algorithm
5	To draw Two Dimensional objects on the screen and perform two dimensional transformations to them
6	To fill objects using Boundary Fill and Flood Fill approaches
7	To draw the projected view of a Three Dimensional object using perspective and parallel projection and perform three dimensional transformations to it
9	To draw a smooth curve using Bezier curve with designated control points
10	To draw graphical output primitives using OpenGL, perform various transformations and implement algorithms covered in the course

7. Evaluation system and students' responsibilities

Internal Evaluation

In addition to the formal end-semester exam(s), the internal (formative) evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation and presentation etc. The tabular presentation of the internal evaluation is as follows. The components may differ according to the nature of the subjects.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30		
Attendance & Class Participation	10%			
Assignments	20%			

Presentations/Quizzes	10%		Semester-End examination	50
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	20%			
Lab Report/Project Report	30%			
Practical Exam/Project Work	30%			
Viva	20%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student requirements:

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the semester-end examination. Failing to get such a score will be equated with NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End- Semester examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. Failure of a student to attend a formal exam, quiz, test, etc. won't qualify him/her for re-exam. Students are required to complete all the requirements defined for the completion of the course

8. Prescribed Books and References

Prescribed Text Book

Donald Hearn and M. Pauline Baker: Computer Graphics C Version, Prentice-Hall.

Reference Books

Donald Hearn and M. Pauline Baker: Computer Graphics with OpenGL, Prentice-Hall.

James D. Foley, Andries van Dam, Steven K. Feiner, John F. Hughes, Computer Graphics: Principles and Practice in C, Addison-Wesley

Mason Woo, Jackie Neider, Tom Davis, Dave Shreiner, "OpenGL Programming Guide", The Official Guide to Learning OpenGL, OpenGL Architecture Review Board, LPE Pearson Edition Asia

Pokhara University and Faculty of Science Technology	
Course Code: CMP 335(3 Credit)	Full Marks: 100
Course Title: Data Communication (3-0-2)	Pass Mark: 45
Nature of the Course: Theory and Practical	Total Lectures: 45 hours
Level: Bachelor/ Year: III/ Semester: VI	Program: BE

1. Course Description:
Data Communication is a fundamental aspect of computer engineering and networking. This course provides students with a comprehensive understanding of the principles, protocols, and technologies used to transmit data between computers, devices, and networks. It explores the theoretical and practical aspects of data communication.
2. General Objectives:
a) To acquire the knowledge and concept of Data Communication. b) To recognize the requirements of digital devices to exchange data. c) To realize the basics of switching and networking.
3. Methods of Instructions:
• Lecture, and practical.

4. Course Contents
Unit 1: Introduction (2 hrs)
• Evolution of Data Communication systems • Analog and Digital Data Transmission, Data Communication Terminology • Standards Organizations, Applications
Unit 2: Data Transmission (3 hrs)
• Data Transmission Techniques - Parallel Transmission, Serial Transmission (Synchronous, Asynchronous and Isochronous Communication), Modes of Data Transmission • Line Configuration, Bit Rate/ Baud rate, Transmission Channel, Data Rate Limits - Shannon Capacity Theorem and Nyquist Bit Rate, RS-232C (DTE-DCE, DTE-DTE)
Unit 3: Signals and Systems (6 hrs)

• Signals and their classification: Periodic and non-periodic signals; Deterministic and Random signals; Energy and Power signals; Continuous and Discrete time signals • Basic Elementary Signals - Unit Step Signal, Ramp Signal, Impulse Signal, Sinusoidal Signal, Signum Signal • System - Continuous and Discrete time system • Basic system properties: Linearity, Causality, Stability, Static & Dynamic, and Time Invariance, Introduction to LTI System
Unit 4: Overview of Data Communication Networking and Protocols (4 hrs)
• Network Types, Topology • OSI layers and Functions, TCP/IP layer, Local Area Networks (LAN) Architecture, LLC/MAC & Routing • IEEE Standards, Ethernet (Aloha, CSMA), Wide Area Networks (WAN): X.25, Frame Relay, ATM
Unit 5: Transmission Media (5 hrs)
• Electromagnetic Spectrum for Telecommunication • Type of Propagation • Guided Transmission Media: Twisted Pair Cable, Coaxial Cable, Optical Fiber, Characteristics of Unguided Communication Bands, Antennas • Unguided Transmission Media: Terrestrial Microwave, Satellite Communication, VSAT, and Cellular Telephony
Unit 6: Impairments, Error handling and Compression Techniques (6 hrs)
• Attenuation & Distortion, Delay Distortion, Noise & Types, interference, crosstalk • Types of error & its Detection and Correction Methods • Data Compression, Lossless Compression - Run Length Coding, Dictionary Coding and Huffman Coding • Lossy Compression - Predictive Coding and Transform Coding
Unit 7: Data Link Control and Protocol (5 hrs)
• Framing • Flow Control: Stop - & - Wait, Sliding Window, Error Control: Automatic Repeat Request (ARQ), Stop-and Wait ARQ, Sliding Window (ARQ) • HDLC protocol • Point-to-Point protocol
Unit 8: Multiplexing & Switching (5 hrs)
• Multiplexing types and Application • Multiplexing Vs Non-Multiplexing • The Telephone System: Analog services and its Hierarchy • Digital services and Hierarchy Circuit Switching, Packet Switching, Message Switching, and Private Branch Exchange

Unit 9: Data Encoding & Modulation (9 hrs)

- Line coding – Unipolar, Polar and Bipolar signaling,
- Digital Modulation Techniques - Amplitude, Frequency, and Phase Shift Keying.
- Analog to Digital Conversion - Pulse Code and Delta Modulation.
- Analog Modulation Techniques -Amplitude, Frequency, and Phase Modulation
- Multilevel Modulation-QPSK, QAM
- Introduction to Modem

5. List of Tutorials

SN	
1.	X
2.	x
3.	x

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester End	50
Attendance &Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance &Class Participation	10%			

Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			

6. List of Practicals	
SN	
1.	Signal Analysis using MATLAB (Maximum 3 Labs)
2.	Implementation of Error Detection Techniques.
3.	Simulated simple PCM coder that converts samples into a digital code
4.	Amplitude Modulation and Demodulation
5.	Frequency Modulation and Demodulation
6.	Simulated Error Control Coding Techniques

7. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Total Internal		50		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References
Text Books: 1. William Stallings, Data and Computer Communications, Fifth edition 2. Behrouz Forouzan, Introduction to Data Communications and Networking

References:

1. Oppenheim, Signals and System
2. U. D. Black, Data Communications and Distributed Networks

Pokhara University
Faculty of Science and Technology

Course Code: CMP 222

Full Marks: 100

Course Title: Database Management System (3-1-3)

Pass Marks: 45

Nature of the Course: Theory and Practical

Total Lectures: 45 hours

Level: Bachelor

Program: BE

1. Course Description

This course is designed to encompass the fundamental concepts of database management system. These concepts include the aspects of database design, database languages and database system implementation. This course presents the introductory concepts of database security, query processing and optimization, transactions and their concurrency control and recovery policies. This course also introduces the emerging new trended databases such as NoSQL and Blockchain databases. After completion of this course, students can design and implement a database system to develop a software application.

2. General Objective

- To acquaint the students with the knowledge of relational database design using ER Model.
- To develop the skills in students to design normalized relational database required for a specified application.
- To acquaint the students with the knowledge of database security, query processing and optimization, files and record organizations, transaction, concurrency control, data recovery mechanisms.
- To acquaint the students with basic concepts of NoSQL databases and Blockchain Databases.

3. Methods of Instruction

- Lecture, Tutorial/Discussion/Readings, Practical works and Project works.

4. Contents in Detail

Specific Objectives	Contents
2 Familiarize with basic concepts of database systems, and its architecture.	Unit 1: Introduction [4 hrs] 1. Database Management System purpose and applications, 2. Database Systems vs File Systems 3. View of Data- Data Abstraction (Physical, logical and view level, Data Independence) 4. Instances and Schemas, 5. Database Languages (DDL, DML and DCL) 6. Database and Application Architecture- Database System Architecture and Database Application Architecture (two-tier and three-tier)

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- Familiarize with ER Model and Relational Model and relational algebra. - Design the relational database using ER model.	Unit 2: ER and Relational Model [8 hours] 1. Introduction to ER Model: 2. Entity sets, attributes and values, Relationship sets- participation, entity's role, descriptive attributes, degree of relationship set, Mapping Cardinalities, Attributes- simple, composite, single-valued, multivalued, derived, Entity-Relationship (ER) Diagram, Specialization, Generalization, and Aggregation 3. Key- Super key, Candidate key and Primary key, Strong and Weak Entity Sets 4. Introduction to Relational Model 5. Reducing ER diagrams to Relational Schema 6. Structure of Relational Databases, Database Schema, Keys, Schema Diagrams 7. Relational Algebra
Implement and write DDL and DML queries in the SQL.	Unit 3: Structured Query Language [5 hours] 1. Structured Query Language (SQL)- SQL DDL and DML, 2. Basic Structure of SQL Queries, DDL queries, Basic Operations (Rename, String, Attribute Specification in the select clause, order by, where-clause), Set Operations, Null values, Aggregate Functions, 3. Nested Queries, 4. Join Expressions (Natural Join, Join Conditions, Outer Joins), 5. Modification of Database (delete, insert into, update), 6. Views, Stored Procedures
- Apply the integrity constraints to implement database securities. - Normalize the database to a defined normal form.	Unit 4: Relational Database Design [6 hrs] 1. Integrity constraints- Domain Constraints, Entity Integrity Constrains, Referential Integrity Constraints, Assertions and Triggers 2. Features of Good Relational Designs 3. Functional dependencies and Armstrong's Axioms 4. Closure of a Set of Functional Dependencies and Closure of Attribute Sets 5. Database Normalization and Normal Forms- 1NF, 2NF, 3NF and BCNF 6. Denormalization for Performance

• Understand the need of database security. • Understand the role of access control, authorization, views and encryption mechanism to provide database security.	Unit 5: Security [3 hrs] 1. Security and integrity violations 2. Access control 3. Authorization 4. Security and Views
Understand the mechanism of query processing and need of query optimization.	Unit 6 : Query Processing and Optimization [5 hrs] 1. Introduction to Query Processing 2. Equivalence of Expressions 3. Query Cost Estimation 4. Query Optimization
	5. Query evaluation and execution plan
? Familiarize with different file organization methods.	Unit 7 : Storage Management and Indexing [3 hrs] 1. File Organization- fixed length records, variable length records 2. Organization of Records in File- Heap, Sequential and Indexed sequential file organizations. 3. B+ Tree Index Files 4. Hash Indices
? Understand the concepts of transactions, need of atomicity, durability and isolation, serial schedule. ? Familiarize with the need of Concurrency control.	Unit 8 : Transactions and Concurrency Control [4 hrs] 1. Transaction Concepts 2. Transaction Model and State Diagram 3. ACID properties of transaction 4. Serializability- conflict and view serializability 5. SQL Standard Isolation Levels 6. Concurrency Control- Lock-Based Protocols and Graph-Based Protocols
? Understand the recovery algorithms and techniques to protect and recover the data from various failures.	Unit 9 : Crash Recovery [3 hrs] 1. Failure classification 2. Recovery and Atomicity- log records, database modification, concurrency control and recovery, transaction commit, Redo and Undo Transactions using Log, Check Points, 3. Recovery Algorithm Using Log Records- Transaction Rollback, Recovery After a System Crash, Optimizing Commit Processing 4. High Availability Using Remote Backup System

7 Understand the concepts of NoSQL, ODBMS, Distributed Databases	Unit 10 : Emerging Trend in Databases [4 hrs] 1. NoSQL Databases- Characteristics, Categories, Advantages 2. Object Oriented Database and ORM 3. Distributed Databases- 4. Distributed Ledged Technology: Blockchain, Cryptocurrency, Blockchain Properties
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5. Practical Works

Laboratory work of 45 hours per group of maximum 24 students should cover the database design, use of database languages and database system implementation using any relational database management system such as MS SQL or MySQL or Oracle etc. Students should complete the following tasks in laboratory:

SN	Tasks to Complete
1	Introduction to MS SQL (or any RDBMS), its datatypes and its installation.
2	All SQL DDL operations studied in Unit 2 such as creating database, creating tables, delete database, drop table, alter etc.
3	All SQL DML operations studied in Unit 2 such as database modification operation- insert into, delete, update etc.
4	Implementing Join Expressions (Natural Join, Join Conditions, Outer Joins)
5	Implementing Stored Procedures.
6	Illustration and implementation of Views.
7	Implementing Integrity constraints (Domain Constraints, Entity Integrity Constrains, Referential Integrity Constraints)
8	Implementing Assertions and Triggers.
9	Implementation of Database Security and Privileges: Grant and Revoke Commands, Commit and Rollback Commands.
10	Connecting database with connection string using any standard programming language and executing SQL queries.

Students should submit a project work that uses all the knowledge obtained from this course to design and implement a database system for any application that students chose. The students should design the database using ER model and present using ER diagram which are then reduced to relational schema. The students should apply the constraints studied in this course including triggers. The database should be in at least 3NF. The marks for the practical evaluation must be based on the project work submitted by students.

6. List of Tutorials

The various tutorial activities that suit your course should cover all the content of the course to give students a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments or class works to the instructor for evaluation. The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover the content of this course:

A. Discussion-based Tutorials: (3 hrs)

- a. Evolution of database systems- traditional file systems and need of database system (Class discussion)
- b. Semi-Structured Data Model and Object-Based Data Model (Class discussion).
- c. Concurrency Control- Lock-Based Protocols and Timestamp-Based Protocols (Oral Presentation).

B. Problem solving-based Tutorials: (6 hrs)

Suppose you are given a task to design a database of a university or college or library, or any real world system that you know very well. Use ER model to design the database for the system you chose, reduce it into the relational schema, represent in Database diagram, normalize the database, implement the integrity constraints, apply some level of security using Grant and Revoke Commands, Commit and Rollback Commands, and develop a mini project. You can combine this tutorial with your lab project stated in Practical Works section.

C. Review and Question/Answer-based Tutorials: (6 hrs)

- a. Case study on any NoSQL Database.
- b. Case study on using Blockchain Database.
- c. Students ask questions within the course content, assignments and review key course content in preparation for tests or exams.

7. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, internal assessment, lab reports, project works etc. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books

1. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2011). Database system concepts. McGraw Hill.

References

1. Majumdar, A. K., & Bhattacharyya, P. (1996). Database Management Systems. McGraw-Hill.
2. Elmasri, R., & Navathe, S. B. (1994). Fundamentals of Database Systems Benjamin Cummings. Redwood City, CA.
3. Everest, G. C. (1986). Database management. McGraw-Hill, Inc..

Pokhara University and Faculty of ScienceTechnology	
Course Code: CMP XXX (3 Credit)	Full Marks: 100
Course Title: Microprocessor and ALP(L-T-P)	Pass Mark: 45
Nature of the Course: Theory and Practical	Total Lectures: 45 hours
Level: Undergraduate Year: Second Semester: III	Program: B E

1. Course Description:
This course introduces microprocessor architecture and microcomputer system, including memory and input/output interfacing devices .It includes assembly language programming, bus architecture, bus cycle types, I/O systems, memory systems, interrupts. This course explores the inner workings of a microprocessor from the programmer's perspective and several laboratory exercises will be based on a microprocessor utilizing the assembly language.
2. General Objectives:
The main objective of the courses are: • To know architecture and organization and addressing modes of a INTEL Microprocessor (8085/8086). • To Introduce Basic operations, programming and application of Microprocessor. • To design and analyze assembly programming code to use the branching structures, looping structures, flags, stacks, procedures, macros, and interrupts. • To Interface memory and I/O devices with the Microprocessor. • To develop a good understanding of the microprocessor operation and how they can be used as building blocks for automated systems and control applications.
3. Methods of Instructions:
• Lecture • Tutorial • Discussion • Laboratory Work

Content in Detail

Specific Objectives	Content
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□ State and interpret block diagram of a Microprocessor based system and explain the function of each component ,the line of communication Bus	Unit1: Introduction to Microprocessor (3 hrs.) 1.1Historical evolution of INTEL series Microprocessor 1.1 Bus organization of a microprocessor 1.2 Microcontroller and its applications 1.3 Microcomputer 1.3.1 Von Neumann and Harvard Architecture 1.4 Concept of fetch, decode and execution
□ Describe the function of each block of the 8085 microprocessors □ Explain the bus timings in fetching an instruction from memory □ Explain the addressing Mode □ Explain the function of data transfer instruction, Arithmetic instruction, logical and conditional and unconditional instruction. □ Enable to write ALP. □ Generation of control signal	Unit 2: Architectural Details and Programming of 8085 microprocessor [12hrs] 2.1 Internal Architecture and features of 8085 Microprocessor 2.2 Intel 8085 Instruction Set 2.3 Classification of Intel 8085 Instruction 2.3.1 Data Transfer Instruction, 2.3.2Arithmetic Instruction 2.3.3 Logical Instruction, 2.3.4Branching Instruction and 2.3.5Stack, I/O and Machine control Instruction 2.4 Addressing Modes of 8085 and its types 2.5 Instruction Cycle, Machine Cycle and T-State 2.6 Timing Diagram 2.6.1 Opcode fetch Cycle 2.6.2 Memory Read Cycle 2.6.3 Memory Write Cycle 2.6.5 I/O Read Cycle 2.6.5 I/O Write Cycle 2.7 Assembly Language Program 2.7.1 Format of an Assembly Language Instruction 2.7.2 Simple Programs with Arithmetic and Logical Operations ,Conditions and Loops

Interpret the block diagram of an 8086 Microprocessor and explain the function of each component. Understand the concept of memory segmentation and anys of physical addressing and effective addressing Knowledge of real instructions and pseudo instructions Enable to write advance programming in ALP Analyze the concept of modular programming	Unit 3 Architectural Details and programming of 8086 microprocessor [10hrs] 3.1 Internal Architecture and features of 8085 Microprocessor 3.1.1 BIU and EU 3.1.2 Memory Segmentation 3.1.2.1 Segment and Offset Address 3.2 Intel 8086 Instruction Set 3.3 Classification of Intel 8085 Instruction 3.3.1 Data Transfer Instruction, 3.3.2 Arithmetic Instruction 3.3.3 Logical Instruction, 3.3.4 Branching Instruction 3.3.5 String manipulation Instruction 3.3.6 Machine control Instruction 3.5 Addressing Modes of 8086 and its types 3.5 Assembly Language program 3.5.1 Format of an assembly language Instruction 3.5.2 ALP Development Tools: Editor, Assembler, Linker, Debugger, Locator, Emulator 3.5.3 Assembler Directives 3.5.4 DOS Interrupt function(Keyboard and display services) 3.5.5 Simple Programs for Arithmetic, Logical, String Input/Output, Conditions and Loops 3.5.6 Stacks Procedures 3.5.7 Macros program design
Understand the concept on types of	

memory and its bus connection with microprocessor Explain the difference between the □ peripheral mapped and memorymapped technique □ Interface and I/O device to a microcomputer by using logic gates and MSI Chip, such as decoders, latches and buffers. □ Explain the Full Address decoding and partial decoding □ List the element of the 8255 programmable peripheral interface (PPI) and explain its various operating modes and control word □ Explain the process of the Direct Memory Access (DMA) and the	Unit 4 Bus Structure, Memory, I/O and Interfacing [12 hrs.] 4.1 Bus Structure 4.1.1 Synchronous Bus 4.1.2 Asynchronous Bus 4.2 Memory Device Classification and Hierarchy 4.3 Interfacing I/O and Memory 4.3.1 Address Decoding 4.3.2 Unique and Non-Unique Address Decoding 4.3.3 I/O Mapped I/O and Memory mapped I/O 4.3.4 I/O Address Decoding (8085) 4.3.5 Memory Address Decoding (8085) 4.4 Parallel Interface 4.4.1 Modes: Simple I/O, Strobe I/O, Single Handshaking and Double Handshaking 4.4.2 Programmable Peripheral Interface (PPI) 4.4.2.1 Architecture of Intel (8255-PPI) 4.5 Serial Interface
function of various elements of the 8237.	4.5.1 Synchronous and Asynchronous Transmission 4.5.2 Programmable Communication Interface 4.5.2.1 USART (8251) 4.6 DMA and Intel 8237A DMA Controller
• Apply and analyze an Interrupt process and the difference between Maskable and Non Maskable Interrupt • Apply and analyze the instruction EI, DI, and RST and their function in the 8085-interrupt process. • Apply and analyze the concept of interrupt subroutine program • Apply and analyze the concept of interrupt processing • Apply and analyze the features of the PIC 8259	Unit 5 Interrupt Operation [8 Hrs.] 5.1 Interrupt , sources of interrupt and its Classification 5.2 Interrupt Processing Sequence in 8085/8086 5.3 Interrupt Instruction 5.3.1 Interrupts pins and priorities 5.3.2 Interrupt Service procedure 5.3.3 Interrupt Vector Table 5.3.4 Multiple Interrupt Handling 5.3.4.1 Polled and Vectored interrupt 5.3.5 Programmable Interrupt Controller (PIC) Intel (8259) and its Architecture

5 List of Tutorials

The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover all the required contents of this course:

SN	Tutorials
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1.	Timing Diagram of 8085
2.	Calculation of Starting and ending address of memory for 8085
3.	Design an interfacing circuit/Address Decoding to interface different size of memory for 8085
4.	Design with input output interfacing with 8085
5.	Assembly Language Program in 8085 and 8086

6. Practical works (30 hours for a group of maximum 24 students)

SN	Practical
1.	Assembly language program using 8085 and 8086 microprocessor kit or simulator Program should comprise the use of all types of instructions and addressing modes
2.	The programming should include the concept of Arrays and the concept of Multiplications and Division operations on 8085 Microprocessor

5. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester End Examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books:

1. Ramesh S. Gaonkar, "Microprocessor Architecture, Programming and Application with 8085", 5th Edition 2002, Prentice Hall

References:

1. D. V. Hall, "Microprocessor and Interfacing, Programming and Hardware", 2nd Edition 1999, Tata McGraw Hill
4. John Uffenbeck, "Microcomputers and Microprocessors, the 8080, 8085 and Z-80 Programming, Interfacing and Troubleshooting" 3rd Edition 1999, Prentice Hall
5. Walter A. Triebel and Avtar Singh, "The 8088 and 8086 Microprocessors, Programming, Interfacing, Software, Hardware and Applications", 4th Edition 2003, Prentice Hall
6. Barry B. Brey, "The Intel Microprocessors", 8th Edition 2011, Pearson Education

Pokhara University
Faculty of Science and Technology

Course Code: CMP 232
Course title: Operating Systems (3-1-2)
Nature of the Course: Theory and Practical
Level: Bachelor

Full Marks: 100
Pass Marks: 45
Total Lectures: 45 hours
Program: BE

1. Course Description

This course is designed to encompass the fundamental concepts of operating system. These concepts include the system software, internal construct of operating system, run application software and perform various tasks. This course also introduces the emerging new trended operating system for real-time, distributed, cloud, mobile systems. After completion of this course, students can design and develop an operating system for any devices.

2. General Objective

- To acquaint the students with basic concepts of resource allocation and management.
- To acquaint the students with structure of operating systems and their functionality.
- To develop the skills in students to select and design optimal resource allocation schedules.
- To acquaint the students with the knowledge of process and thread, I/Os, Memory, CPU, disk management.
- To acquaint the students with basic concepts of Operating systems new trends such as real-time, distributed, cloud, and mobile systems.

3. Methods of Instruction

Lecture, Discussion, Readings, Practical works and Project works.

4. Contents in Detail

Specific Objectives	Contents
- Familiarize with basic concepts of Operating systems, and its architecture. - Understanding the successive operating system	Unit 1: Introduction [4 hrs.] 1.1 Concept and function of operating systems 1.2 Operating systems concept and functionality - Processes, Files, System Calls, The Shell evolution of 1.3 Operating System Structure - Monolithic System, Layered, Virtual Machines, Client-server. Microkernel and exokernel 1.4 Types and Evolution of Operating Systems
Familiarize with Task,	Unit 2: Processes and Threads [10 hours] Process and threads 2.1 Process

• Implement of resource allocation - Definition, states and transition diagram techniques - Process control block (PCB) • Understanding the mutual - Concurrent and Parallel processes exclusion for resource utilization	2.2 Interprocess Communication and Synchronization - Introduction, Race Condition, Critical Regions and condition, Avoiding critical region: Mutual Exclusion and Serializability,

- Mutual exclusion conditions. mutual exclusion:

	disabling interrupts, lock variable, strict alteration (Dekker's algorithms, Peterson's algorithms), The TSL instruction, sleep and wakeup, producer and consumer problem - Types of mutual exclusion (Semaphore, Monitors, Bounded buffer, Message passing), - Classical IPC Problems (The Dining Philosophers problem, The Readers and Writers problem, The Sleeping Barber Problem) - Serializability: Locking Protocols and Time Stamp Protocols 2.3 Process Scheduling - Basic Concept, Type of scheduling (Preemptive scheduling, Non-preemptive scheduling, Batch, Interactive, Real time scheduling), - Scheduling criteria and performance analysis, scheduling algorithm with examples (First come first served, Shortest-job-first, Roundrobin, Shortest process next, Shortest remaining time next, Real time, Priority fair share, guaranteed, Lottery scheduling) 2.3 Threads - Definitions of Threads, Types of thread process (Single and multithreaded process), Benefits of Multithreading - Threads Models (Many-to-one model, One-to-one Model, Many-to many model) - User Space and Kernel Space Threads, - Difference between Processes and Threads
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❑ Conceptualize the role and working procedure of memory ❑ Familiarizing with virtual memory management ❑ Understanding the page replacement algorithms	Unit 3: Memory Management [10 hours] - Introduction: Storage organization, Memory hierarchy - Storage allocation, Contiguous versus noncontiguous store allocation, logical and physical memory - Fragmentation, fixed partition and variable partition for multiprogramming, Logical versus physical address space - Relocation and Protection - Memory management with swapping: Memory management with bitmaps and linked list, Memory management without swapping, - Contiguous-memory allocation: memory protection, memory allocation, Fragmentation (Inter fragmentation and external fragmentation) - Paging, Structure of page table: Hierarchical page table, Hashed page table, Inverted page table, Shared page table, - Virtual memory- Introduction, Paging, Page Table, Block mapping, Direct mapping, Translation Look Aside Buffers) Demand
	paging, Thrashing - Page replacement, Page replacement algorithms: First-in-first-out, Not recently used, Optimal page replacement, Second chance page replacement, Least Recently used, Clock page replacement, Working set page replacement, WS clock page replacement - Segmentation, Segmentation with paging - Coalescing and Compaction
❑ Conceptualization of Kernel and its role in system software	Unit 4: Kernel [2 hrs] 1. Introduction, Architecture of the Kernel, Types of Kernels 2. Context Switching (Kernel and User mode), Kernel implementation processes

❑ Understand the role of input/output devices ❑ Understand the different approaches for optimal output	Unit 5: Input/Output Management [5 hrs] - Introduction, Interrupts Handlers - Principles of I/O Hardware (I/O Device, Device Controller, Memory Mapped I/O, Direct Memory Access) - Principles of I/O Software (Goals of I/O Software, Polled I/O verses Interrupt Driven I/O, Character User Interface and Graphical User Interface, Device Driver, Device Independent I/O Software, User -space I/O Software, - System Resources: Preemptable and Non-preemptable, Method of handling Deadlocks, Deadlock prevention, - Deadlock avoidance: Banker's Algorithm, Deadlock detection: Resource allocation graph, Recovery from Deadlock - Redundant Array of Inexpensive Disks - RAM Disks
❑ Understand the file and filing mechanism ❑ Familiarize with directory and its management techniques	Unit 6: File Systems [3 hrs] - File and File Organization (Blocking and Buffering, File Descriptor, file Naming, File Structure, File Types, File Attributes, File Operations, File Access Methods) - Directories Management (Single-level directory systems, Hierarchical Directory systems, Path names, Directory operation) - Access Methods: Sequential, Direct, other access methods, Protection: Types of access, Access Control List, - File System Implementation: Contiguous allocation, Linked list allocation, linked list allocation using an Index nodes - Security and Multi-media files
❑ Familiarize with current trends of operating systems	Unit 7: New Trends in Operating System - Concept, character and role of - Real-time Operating System, - Distributed Operating System - Cloud Operating System - Mobile Operating System - Security issues and method of deployment - Memory wall and bottleneck for Operating system

5. Practical Works

Laboratory work of 12 hours should cover the operating system such as MS Windows, Linux, etc. Students should complete the following tasks in laboratory:

1. Installation and user, application management in Windows (current version)
2. Simulation of Process Scheduling Algorithms using C/C++
3. Simulation of Disk Arm Scheduling Algorithms
4. System Administration (user, disk, role, etc.) in any open-source operating system.

6. List of Tutorials

The various tutorial activities that suit your course should cover all the content of the course to give students a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments or class works to the instructor for evaluation. The following tutorial activities of 10 hours as per the convenient of student and faculty.

A. Review and Question/Answer-based Tutorials: (10 hrs)

- a. Case study on any Open-source operating system.
- b. Students ask questions within the course content, assignments and review key course content in preparation for tests or exams.

7. Evaluation System and Students'

Responsibilities Evaluation System

The internal evaluation of a student may consist of assignments, attendance, internal assessment, lab reports, project works etc. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			

Total Internal		50	
Full Marks: 50 + 50 = 100			

Student Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books

- 1) 1. Andrew s. Tanenbaum, "Distributed Operating System", Pearson

References

- 1) Andrew s. Tanenbaum, "Modern Operating System", PHI, 3rd Ed. 2011
- 2) A. Silberschatz, P.B. Galvin, G. Gagne "Operating System Concepts", Wiley, 8th Ed.
- 3) D M Dhamdhere, "System Programming and Operating System"- Tata McGraw-Hill, 20

